



CS201 DSA PROJECT

Reimagine Sentiment Analysis for Airline Reviews Insights

Data Structures for
Polarized Word-Frequency Counts

G2T3 - TEAM TOUCH GRASS



NOV 2025



Problem Statement

Information Overload



**Difficult to Interpret
Large Volumes of Text**

Sentiment Differentiation



**Words Distinguish Positive
from Negative reviews?**

Efficient Computation at Scale



**Slow and memory-heavy
as Sentiment Data Scales**

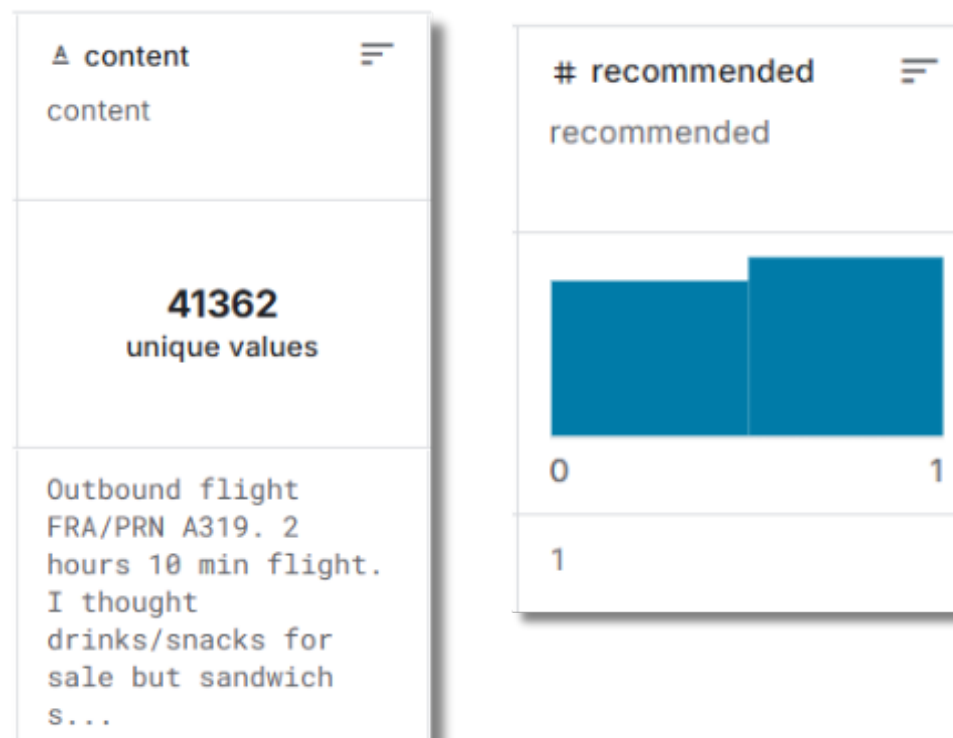
Exploring Data Structures in Sentiment Analysis

Data Tokenization Pre-Processing

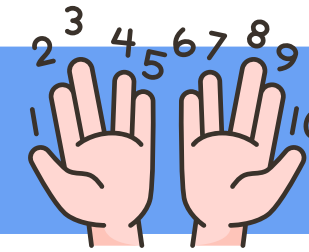


SkyTrax dataset

- Pronoun Filtering
- For a given airline, **split reviews** by their **"recommended"**



Counting the Tokens/Words



- Count Token Frequencies
- Extract Top-K per class

```
[BST] Time (median of 5): 23 ms | Allocated (median): 330 KB
[GOOD] Top 10 most common words in POSITIVE reviews:
1. Word: about      Count: 136
...
[BAD] Top 10 most common words in NEGATIVE reviews:
1. Word: hour       Count: 679
...
```

- Interactive Comparisons

```
=====
COMPARISON RESULTS FOR: "DELAY"
Sentiment: NEGATIVE reviews
Total airlines with this word: 281
Minimum reviews for reliability: 50
=====

TOP 5 Airlines with HIGHEST weighted score for "delay":
(Weighted score considers both frequency % and review count)

Rank Airline      Reviews    Count    Percent    Score
-----
1  sunwing-airlines  286       407    142.31%    349.78
2  allegiant-air     351       375    106.84%    272.07
3  spirit-airlines    786       623     79.26%    229.54
4  norwegian         178       175     98.31%    221.49
5  flybe             227       205     90.31%    212.94

BUILD SUCCESSFUL in 1m 8s
3 actionable tasks: 1 executed, 2 up-to-date
```

Experimental Results



- Analyze Time Complexity and Space Complexity

BST Implementation Test	Red-Black Tree Implementation Test	Arraylist Implementation Test	Map Implementation Test
Time (median of 5): 23 ms Allocated (median): 330 KB	Time (median of 5): 22 ms Allocated (median): 370 KB	Time (median of 5): 53 ms Allocated (median): 2002 KB	Time (median of 5): 23 ms Allocated (median): 3274 KB
[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:
1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524
2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330
3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289
4. Word: airline Count: 258	4. Word: airline Count: 258	4. Word: airline Count: 258	4. Word: airline Count: 258
5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221
6. Word: all Count: 179	6. Word: all Count: 179	6. Word: all Count: 179	6. Word: all Count: 179
7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153
8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144
9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142
10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137
[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:
1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117
2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947
3. Word: airline Count: 904	3. Word: airline Count: 904	3. Word: airline Count: 904	3. Word: airline Count: 904
4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690
5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679
6. Word: delay Count: 623	6. Word: delay Count: 623	6. Word: delay Count: 623	6. Word: delay Count: 623
7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557
8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544
9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541
10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533

- 5x, 10x, 15x csv to measure asymptotic complexity

```
# --- Configuration ---
# Set the path to your input CSV here. If your
input_csv = "airline.csv" # <-- change this to
# Replication factors you want to output
factors = [10, 100, 1000]
```


Demo



=====	=====	=====	=====
BST Implementation Test	Red-Black Tree Implementation Test	ArrayList Implementation Test	Map Implementation Test
=====	=====	=====	=====
Time (median of 5): 23 ms Allocated (median): 330 KB	Time (median of 5): 22 ms Allocated (median): 378 KB	Time (median of 5): 53 ms Allocated (median): 2602 KB	Time (median of 5): 21 ms Allocated (median): 1274 KB
[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:	[GOOD] Top 10 most common words in POSITIVE reviews:
-----	-----	-----	-----
1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524	1. Word: flight Count: 524
2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330	2. Word: spirit Count: 330
3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289	3. Word: seat Count: 289
4. Word: airlin Count: 258	4. Word: airlin Count: 258	4. Word: airlin Count: 258	4. Word: airlin Count: 258
5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221	5. Word: time Count: 221
6. Word: all Count: 170	6. Word: all Count: 170	6. Word: all Count: 170	6. Word: all Count: 170
7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153	7. Word: fly Count: 153
8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144	8. Word: bag Count: 144
9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142	9. Word: check Count: 142
10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137	10. Word: pai Count: 137
[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:	[BAD] Top 10 most common words in NEGATIVE reviews:
-----	-----	-----	-----
1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117	1. Word: flight Count: 2117
2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947	2. Word: spirit Count: 947
3. Word: airlin Count: 904	3. Word: airlin Count: 904	3. Word: airlin Count: 904	3. Word: airlin Count: 904
4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690	4. Word: seat Count: 690
5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679	5. Word: hour Count: 679
6. Word: delay Count: 623	6. Word: delay Count: 623	6. Word: delay Count: 623	6. Word: delay Count: 623
7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557	7. Word: fly Count: 557
8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544	8. Word: bag Count: 544
9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541	9. Word: time Count: 541
10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533	10. Word: get Count: 533

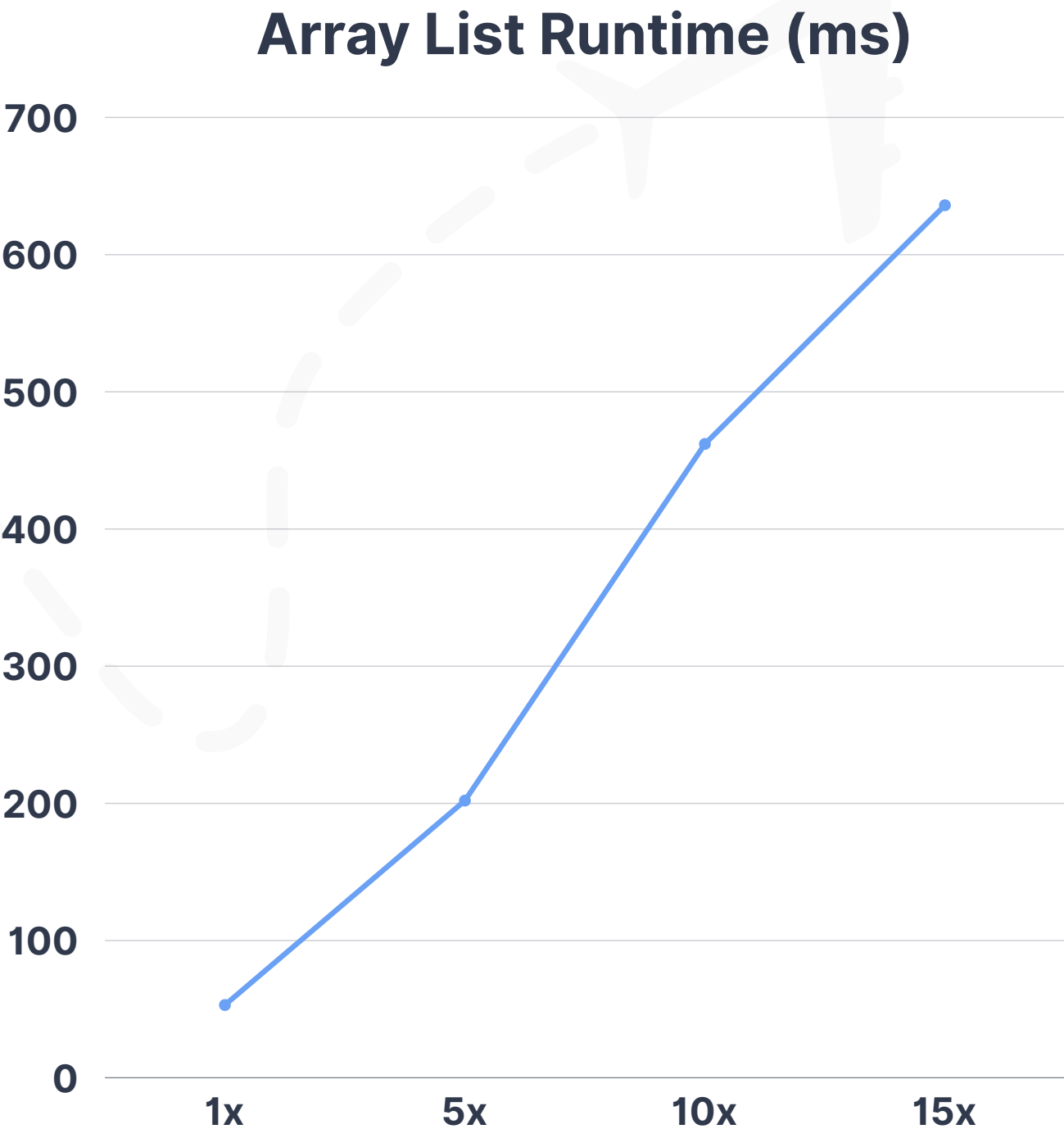
Array List

- Time Complexity: $O(N \times M \times \log(N \times M))$
- Space Complexity: $O(N \times M)$

Where:

- N = Number of reviews
- M = Average words per review

Empirical Results	1x	5x	10x	15x
Runtime (ms)	53	202	462	636
Memory (MB)	2.602	1.235	22.083	32.224



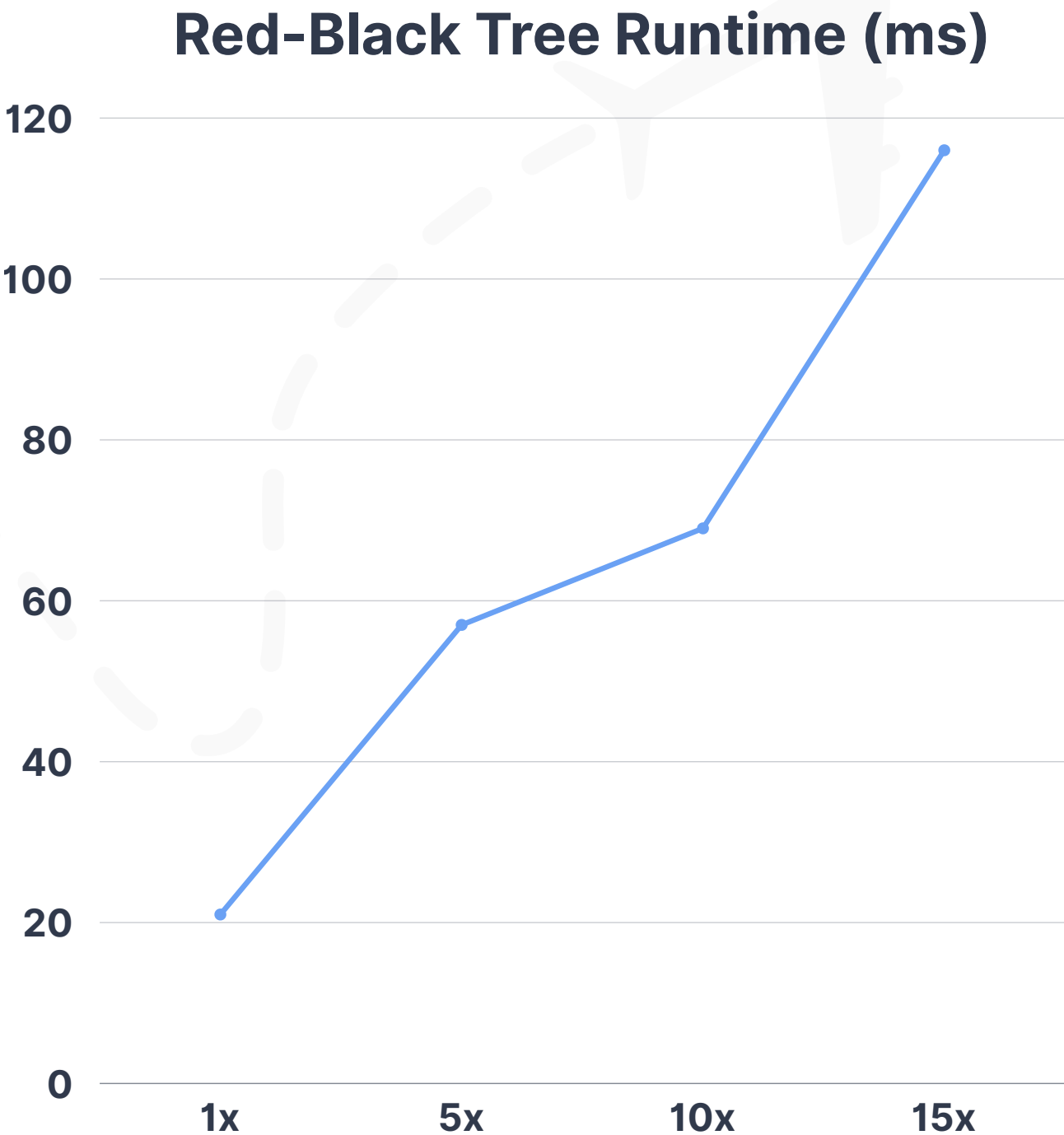
Map

- Time Complexity: $O(N \times M + W \log W)$
- Space Complexity: $O(W)$

Where:

- N = Number of reviews
- M = Average words per review
- W = Unique words total

Empirical Results	1x	5x	10x	15x
Runtime (ms)	21	57	69	116
Memory (MB)	1.274	4.657	9.994	15.730



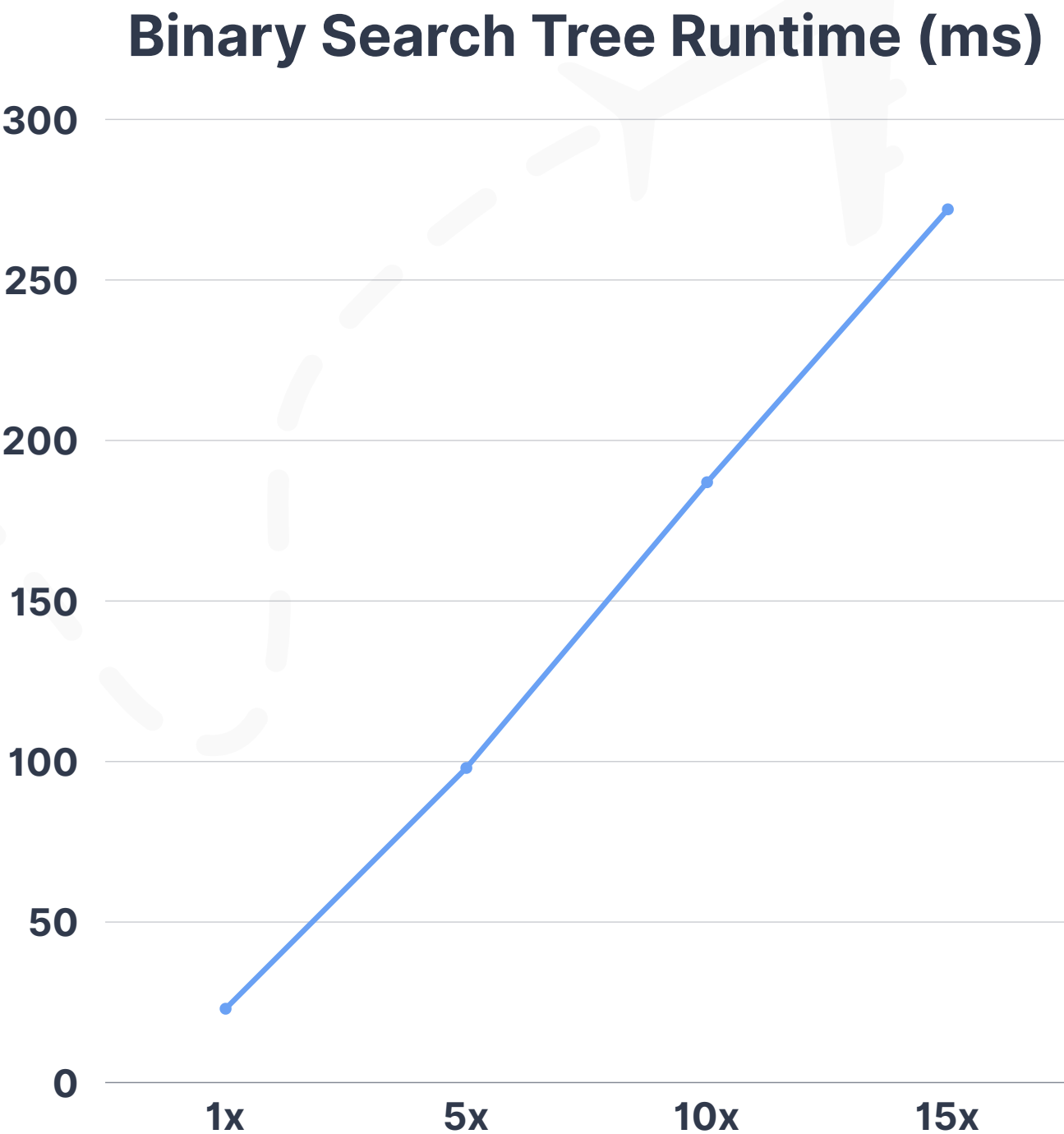
Binary Search Tree

- Time Complexity: $O(N \times M \times \log(N \times M))$
- Space Complexity: $O(N \times M)$

Where:

- N = Number of reviews
- M = Average words per review

Empirical Results	1x	5x	10x	15x
Runtime (ms)	23	98	187	272
Memory (MB)	0.33	0.33	0.33	0.33



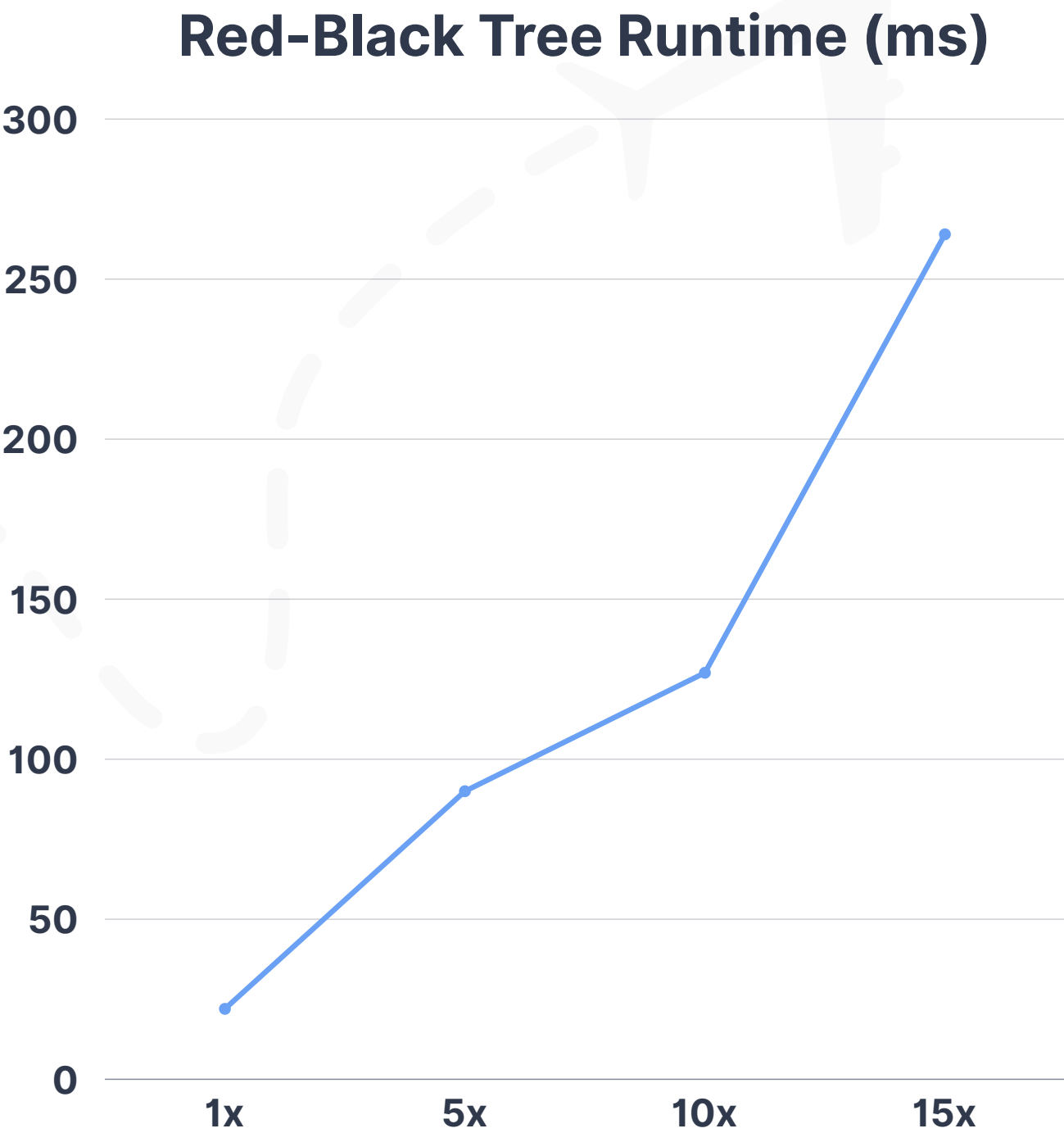
Red-Black Tree

- Time Complexity: $O(N \times M \times \log(W))$
- Space Complexity: $O(W)$

Where:

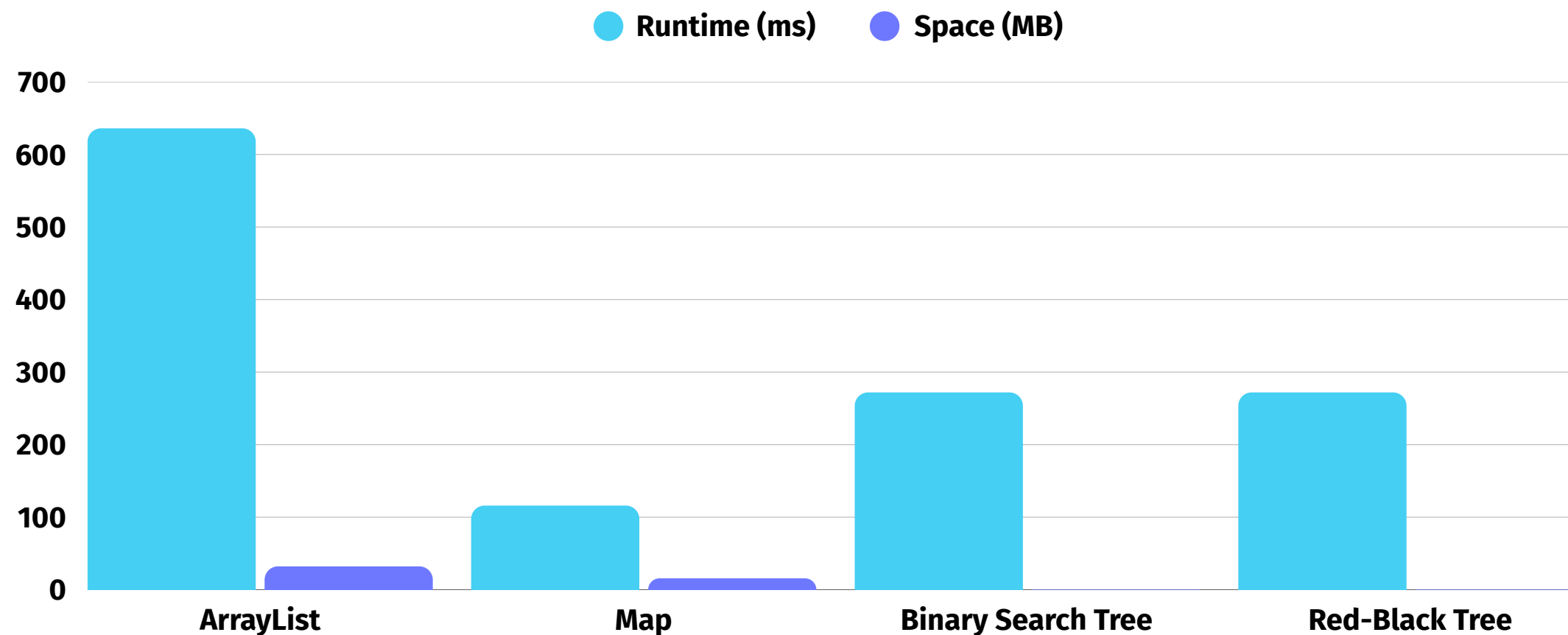
- N = Number of reviews
- M = Average words per review
- W = Unique Words

Empirical Results	1x	5x	10x	15x
Runtime (ms)	22	90	127	264
Memory (MB)	0.378	0.378	0.378	0.378



Empirical Results: Implementation Advantage

Performance Metrics for Data Structures (ADT)



• For **Top-K** filtering

3 Key Insights



Trees uses least and constant space across datasets



Map is the clear winner on speed (runtime)



Recommendation:
Maps for static reviews, RBT for streaming/live-reviews